

VAIBHAV SHARMA

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SUMMARY

Full-stack developer with a backend inclination and a standing interest in security. Currently building production systems end-to-end – from API design through incident response as backend engineer at a seed-stage startup; Open-sourced Arachnode, a 7-microservice distributed platform (26 stars, 47 forks), and maintain it as Project Admin reviewing pull requests and closing issues from outside contributors.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, C, SQL
Backend: Spring Boot, Spring Security, FastAPI, Node.js, Express, REST API Design, JWT, RBAC
Frontend: React, Next.js, Tailwind CSS
Data & Infra: PostgreSQL, MySQL, Redis Streams, MongoDB, Docker, Nginx, AWS EC2
Security: Seccomp, Linux Namespaces, Capability Dropping, rlimits
Testing & Tools: testcontainers (unit/integration/contract/e2e), Git, GitHub Actions, Claude Code, Cursor

INTERSHIPS

Backend Engineering Intern

Jun 2026 – Present

LazyStudents.in

Remote

- Diagnosed a silent multi-step execution failure in the Workflows engine; applied `SELECT FOR UPDATE SKIP LOCKED` for safe job recovery.
- Designed and shipped an NL-to-workflow AI orchestration layer (Lazy Command) using Gemini; built a complexity-based intent router, a typed JSON plan compiler, and dual execution paths covering 100+ platform tools.
- Led a 4-person intern team; drove QA remediation across 40+ production tools and reviewed 23+ PRs.
- Rebuilt a broken webhook inspection tool end-to-end; surfaced an IDOR, login timing oracle, and Next.js build-time auth bypass in a concurrent security audit.

AI/ML Backend Intern

Sep 2025 – Nov 2025

Infosys Springboard

Virtual

- Orchestrated 3 heterogeneous ML models (MusicGen, fine-tuned GPT, YAMNet) behind a Flask API layer for an AI music generation platform; implemented model preloading and graceful fallback routing that cut initialization latency **45%** (2.2s → 1.2s) over 500+ requests.
- Designed error-handling middleware ensuring inference continuity under model failures; persisted metadata and hashed credentials in MongoDB.

PROJECTS

Arachnode — Automated Job Discovery Platform | *Python, FastAPI, Redis Streams, Docker*

- **Problem:** Manually tracking job postings and cold-emailing recruiters was unscalable during placement prep.
- **Solution:** Architected a self-hosted, event-driven platform across **7 microservices** on Redis Streams consumer groups (chosen over Kafka for lower ops overhead at single-node scale), with a 4-tier test suite (unit, integration via testcontainers, contract, e2e) and scheduler-driven OSINT enrichment + AI-drafted outreach.
- **Outcome:** The automated discovery-to-draft cycle now eliminates hours of manual job-search effort per day – and now open-sourced with **26 stars, 47 forks, 143 commits**; serve as Project Admin reviewing **25+ PRs** and mentoring GSSoC/ELUSOC contributors.

Appraisal Management System | *Java, Spring Boot, Spring Security, PostgreSQL*

- **Problem:** College's faculty appraisal process was entirely paper-based, with no audit trail or consistency.
- **Solution:** built a state-machine-driven workflow backend (Draft → Submitted → HOD Reviewed → Finalized → Locked) with JWT + RBAC across 4 roles, a full audit trail per transition, and server-side validation of scoring weights.
- **Outcome:** 13 Flyway migrations applied cleanly; Dockerized with env-injected credentials and auto-generated OpenAPI docs. College administration requested integration into the internal ERP — a projected 50–60% reduction in manual appraisal effort — pending a rebuild in Angular + Node.js to match the ERP's stack.

Offline AI Assistant Shell | *C, llama.cpp, Seccomp, Linux Namespaces*

- **Problem:** I wanted local, air-gapped AI command assistance without exposing the host system to an untrusted AI process.
- **Solution:** built a sandboxed C shell integrating TinyLlama-1.1B via llama.cpp behind a four-layer security model — Seccomp syscall whitelisting, namespace isolation, capability dropping, and rlimit caps.
- **Outcome:** verified via MD5 model-integrity checks and strace-based syscall auditing; runs at 10–20 tok/s on CPU within ~1–1.5GB memory.

VulnGraph — Commit-Level Security Regression Detector | *FastAPI, React, NetworkX, Python AST*

- **Problem:** snapshot-based SAST tools (SonarQube, CodeQL) can flag a vulnerability but can't say which commit introduced it, or whether a recent change quietly removed a validation guard.
- **Solution:** built a program-graph engine that diffs entry-to-sink attack paths between consecutive commits and assigns each commit a deterministic Risk Factor (0–1) based on new paths opened and guard strength lost, using git-diff-scoped incremental analysis so large repos re-scan in seconds.
- **Outcome:** catches regressions no snapshot-based tool detects; deterministic scoring makes it usable as an automated CI/CD gate — validated with Mann-Whitney U testing against CVE-linked Java repositories.

Additional builds: HPMS — Hostel Permission Management System (Node.js/Express/Prisma — 5-role QR-based gate workflow, full SRS proposed to college administration for institutional deployment) • Academic Timetable Scheduling Engine (Java/Spring Boot — greedy constraint-propagation scheduler enforcing 8 hard constraints, generates conflict-free timetables in under 2 minutes) • Air Notepad (OpenCV/MediaPipe gesture-drawing tool — LinkedIn post reached 176,000+ impressions)

LEADERSHIP & COMMUNITY

Project Admin — Arachnode

Feb 2026 – Present

GSSoC / ELUSOC / Aperture 3.0

- Review and merge **75+** contributor PRs, enforce API contract standards, maintain CI/GitHub Actions pipelines, closed **30+** issues, and mentor first-time open-source contributors; merged 2 PRs into the GitCanvas project under Aperture 3.0.

President, Coders Club

Oct 2025 – Present

MITE Campus

- Organized **Hack n' Seek**, a hybrid technical event with **700+** online and **180+** on-site participants across multiple colleges; Delivered **10+** workshops on Git, DSA, and backend architecture to 60+ students.

EDUCATION

Mangalore Institute of Technology and Engineering

Sep 2023 – Jun 2027

B.E., Computer Science and Engineering; CGPA: 9.37

Karnataka, India

Competitive Programming: 150+ LeetCode problems • 100+ TUF/Striver SDE Sheet problems

CERTIFICATIONS

Oracle Cloud Infrastructure

DevOps Professional

Nov 2025

Data Science Professional

Oct 2025

Generative AI Professional

Oct 2025

Generative AI Foundations Associate

Sep 2025

Google Cloud / Coursera

Cloud Security Engineer Specialization

Jul 2024

Security in Google Cloud Specialization

Sep 2024

Google Cybersecurity Professional Certificate

Dec 2023